1. Item 2: Identify the key variables in a model

- **Highly relevant**: The question explicitly provides several key variables:
  - Distance: 9 km uphill, 9 km downhill, total 18 km
  - Speed: 1.5 km/h uphill, 3 km/h downhill (twice the uphill speed)
  - Time constraint: Must return by 8 pm
- **Analysis**: The first step in modeling is to identify and define all relevant variables.

2. Item 10: Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem

- **Highly relevant**: It is necessary to establish the relationships between speed, time, and distance: $t_{\text{up}} = \frac{9}{1.5}$, $t_{\text{down}} = \frac{9}{3}$, total time $T = t_{\text{up}} + t_{\text{down}}$
- **Analysis**: Expressing physical relationships through mathematical formulas is a core step in modeling.

3. Item 4: Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

- **Highly relevant**: Students must understand the logical structure of the task, such as "different speeds for uphill and downhill," "latest return time," and "calculating the latest start time."
- **Analysis**: This is a prerequisite for understanding the problem structure and translating it into a mathematical model.

4. Item 7: Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

- **Relevant**: The walking process is decomposed into two phases: "uphill" and "downhill," with time calculated separately for each before combining.
- **Analysis**: Decomposing complex actions into simpler stages is a common simplification strategy in modeling.

5. Item 6: Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

- **Relevant**: The hiking trip is modeled as a "constant speed motion" problem, ignoring details such as variable speeds and the distribution of rest stops.
- **Analysis**: This is a key step in abstracting a real-world scenario into a mathematical problem.

6. Item 8: Recognize aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures

- **Relevant**: This problem is essentially an application of a "travel problem" or "speed-time-distance relationship," which is a classic mathematical model.
- **Analysis**: Recognizing the type of problem helps in quickly applying appropriate mathematical knowledge.

7. Item 12: Identify the constraints, assumptions, simplifications in a mathematical model

- **Weakly relevant**: Assumptions include constant speeds for uphill and downhill, with meal and rest times already factored in. Factors such as weather or changes in physical condition are ignored.
- **Analysis**: The reasonableness of these assumptions can be discussed during modeling reflection.

1. 第 2 项：Identify the key variables in a model

- **强相关**：题目明确给出了多个关键变量：
  - 路程：上山 9 km，下山 9 km，总路程 18 km
  - 速度：上山 1.5 km/h，下山 2 倍即 3 km/h
  - 时间约束：最晚返回时间 8 pm
- **分析**：建模的第一步就是识别并定义所有相关变量。

2. 第 10 项：Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem

- **强相关**：需要建立速度、时间与路程的关系式：
  $t_{\text{up}} = \frac{9}{1.5}$，$t_{\text{down}} = \frac{9}{3}$，总时间 $T = t_{\text{up}} + t_{\text{down}}$
- **分析**：通过数学公式表达物理关系，是建模的核心步骤。

3. 第 4 项：Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

- **强相关**：学生需要理解“上山与下山速度不同”“最晚返回时间”“需计算最晚出发时间”等任务逻辑。
- **分析**：这是理解问题结构并转化为数学模型的前提。

4. 第 7 项：Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

- **相关**：将行走过程分解为“上山”和“下山”两个阶段，分别计算时间，再合并。
- **分析**：分解复杂动作为简单阶段是常见的建模简化策略。

5. 第 6 项：Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

- **相关**：将徒步行程建模为“匀速运动”问题，忽略实际行走中的变速、休息点分布等细节。
- **分析**：这是将现实场景抽象为数学问题的关键一步。

6. 第 8 项：Recognize aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures

- **相关**：该问题本质上是“行程问题”或“速度-时间-路程关系”的应用，属于经典数学模型。
- **分析**：识别问题类型有助于快速调用合适的数学知识。

7. 第 12 项：Identify the constraints, assumptions simplifications in a mathematical model

- **弱相关**：假设上下山速度恒定，且已包含用餐与休息时间；忽略天气、体能变化等因素。
- **分析**：在建模反思中可讨论这些假设的合理性。